

V-SPACE FPGA Build Challenge 2026

Submission Guidelines

Dear Participants,

To ensure fair evaluation and proper documentation of your work, every team must maintain a structured GitHub repository and submit a detailed project report for all experiments completed during the FPGA Build Challenge.

Deliverables

Each team must submit:

- 1. Project Documentation (PDF)**
- 2. GitHub Repository**
- 3. Demonstration Video**
- 4. FPGA Project Files**

All submissions must be uploaded before the submission deadline.(23.09.2026)

Competition Requirement

Each team must complete:

Difficulty	Number of Experiments
Beginner	2
Intermediate	2
Advanced	1

Total:

5 FPGA Projects

Teams are free to choose experiments according to the capabilities of their allocated FPGA board.

GitHub Repository Structure

Every team must create ONE GitHub repository.

Repository Name Format:

FPGA-Build-Challenge-TeamLeader name

Repository Structure

FPGA-Build-Challenge-TeamName/

```
|
|├── README.md
|
|├── Experiment-1-Beginner/
|    ├── Documentation/
|    ├── RTL/
|    ├── Testbench/
|    ├── Simulation/
|    ├── Images/
|    └── Video_Link.txt
|
|├── Experiment-2-Beginner/
|    ├── Documentation/
|    ├── RTL/
|    ├── Testbench/
|    ├── Simulation/
|    ├── Images/
|    └── Video_Link.txt
|
|├── Experiment-3-Intermediate/
|
|├── Experiment-4-Intermediate/
|
|├── Experiment-5-Advanced/
|
|└── Final_Report/
     └── TeamName_Final_Report.pdf
```

README.md Requirements

The repository home page must contain:

Team Information

Team Name

Team Members

Registration Numbers

Selected FPGA Board

Project Summary

Provide a brief description of all five experiments.

Example:

Experiment 1:

Traffic Light Controller

Experiment 2:

UART Transmitter

Experiment 3:

Digital Clock

Experiment 4:

SPI Communication

Experiment 5:

RISC-V Soft Processor

FPGA Board Used

Example:

Board:

DE10-Lite

FPGA:

Intel MAX10

Documentation Format(PDF)

Each experiment must contain a report including:

1. Objective

What problem is being solved?

2. Block Diagram

System architecture diagram.

3. RTL Design

Explain modules used.

Include:

- Verilog code structure
- Module hierarchy

4. Simulation Results

Include:

- Waveforms
- Testbench outputs

5. Hardware Implementation

Include:

- FPGA screenshots
- Resource utilization (optional)

6. Applications

Real-world usage.

7. Conclusion

Learning outcomes and future scope.

RTL Folder

Must contain:

RTL/

|

├─ top_module.v

├─ submodule1.v

├─ submodule2.v

└─ constraints.sdc/xdc

Testbench Folder

Testbench/

|

└─ tb_top.v

Simulation Folder

Include:

Simulation/

|

├─ waveform.png

├─ transcript.txt

└─ simulation_report.pdf

Images Folder

Must include:

Images/

|

├─ block_diagram.png

├─ rtl_schematic.png

├─ board_setup.jpg

└─ hardware_output.jpg

Video Submission

Upload video to:

- Google Drive

Add the link inside:

Video_Link.txt

Video should include:

Introduction

Problem Statement

What is the experiment?

Design Explanation

Architecture and RTL code in software

Demonstration

Hardware working (setup of hardware + connection with hardware)

Conclusion

Evaluation Focus

Judges will primarily evaluate:

Criteria	Weightage
Technical Correctness	40%
Innovation	15%
Documentation Quality	15%
Video Presentation	15%
FPGA Implementation	15%